MODELING CONVENTIONS

Governing provision

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Controls, Exceptions, and Version Administration

Modeling Conventions Memo identifies its authority, effective date, scope, defined terms, mandatory controls, evidence owners, approvers, escalation paths, exceptions, monitoring, and version history.

An exception requires written rationale, compensating controls, approval, expiry, and continuing monitoring; a prior exception does not create a standing waiver.

Superseded drafts remain available for history but do not control a current transaction merely because they contain later edits or more detailed examples.

Section 1. Purpose, Authority, and Application

This memorandum states the measurement and modelling conventions that every model, schedule and computation prepared at the firm must apply. Issue 4, revision 3 was approved on 2026-03-24 and applies to work performed on and after 2026-04-06. It is issued under the authority of the head of valuation and binds every preparer, reviewer and approver of a model, including advisers preparing a model for the firm's use.

A computation that departs from these conventions is wrong for the firm's purposes even where it is defensible in the abstract, because the value of a convention is that every reader can reproduce the result without asking the preparer what was done. A preparer who believes a convention is inappropriate for a particular calculation must obtain a written departure from the head of valuation before preparing the calculation, not after.

This memorandum states conventions and the location of the assignments that apply them. Where a calculation requires a precision, a rounding direction, a day-count basis, a rate source, a tolerance or an iteration ceiling, the applicable value is taken from the schedule this memorandum names for that purpose or from the executed instrument that governs the quantity. No such value appears in the body of this memorandum, and a preparer who finds one in a copy of it should treat the copy as unofficial.

Every model output issued for use outside the preparer's own working file must declare the conventions it applied and the schedule edition it took its assignments from. An output without that declaration is treated as having applied no convention at all and is returned unread; the reviewer records the return so that the frequency of the failure is visible.

Section 2. Model Classes, Ownership, and the Model Register

Every model is assigned to a class on the model register, and every class has one named owner. The owner is accountable for the model's inputs, its calculation chain, its documentation and its retained versions, and may not delegate accountability by circulating the file to another preparer. Where ownership changes, the register records the date and the outgoing owner certifies the state of the model at handover.

A model may be used to support a decision only where the register records it as approved for decision use. Approval is given by the reviewer named for the class after the controls in section 11 have been evidenced, and lapses on any structural change to the model. A model in reference or illustrative use may not be promoted to decision use by the preparer's assurance that it has not changed materially.

A model supplied by an issuer or a third party is retained as received and is never edited. Where the firm needs to compute from it, the computation is performed in a separate model that cites the supplied file as an input; this keeps the boundary between what the issuer asserted and what the firm derived visible on the face of the record.

Register 2.4 - Model classes, ownership and permitted use

Class	Model type	Owner	Approving reviewer	Version retention	Permitted use
M1	Transaction structuring model	Deal analytics	Head of valuation	Every version retained	Decision use
M2	Capital structure and conversion model	Deal analytics	Transaction counsel	Every version retained	Decision use

Class	Model type	Owner	Approving reviewer	Version retention	Permitted use
M3	Distribution and proceeds model	Deal analytics	Head of valuation	Every version retained	Decision use
M4	Reserve computation model	Reserve analyst	Chief financial officer	Every version retained	Decision use
M5	Return and performance model	Fund controller	Chief financial officer	Quarterly versions retained	Reporting use
M6	Valuation support model	Valuation team	Head of valuation	Every version retained	Decision and reporting use
M7	Operating forecast supplied by an issuer	Sponsoring partner	Deal analytics	As received, unaltered	Reference only
M8	Sensitivity or scenario overlay	Model owner of the base model	Head of valuation	Retained with the base model	Illustrative only
M9	Ad hoc analysis prepared for a single question	Preparer	Senior reviewer	Retained with the question	Not for reuse

Section 3. Measurement Dates and Period Conventions

Every quantity a model reports must be reported as at a stated date or for a stated period, and the date or period must appear on the face of the output rather than in the file name. A figure reported without its date is not a measurement and may not be carried into another model, a register or a report.

The measurement date applicable to a computation is the date the governing document or the applicable measurement calendar designates for it. This memorandum does not designate a date for any computation; where two documents would designate different dates, the preparer records both and escalates rather than selecting one, and the escalation identifies the two documents and the dates they designate.

A period is closed at its stated end and is not extended to capture a later event, however significant the event. An event occurring after the end of a period is disclosed as a subsequent event and is not folded into the period's figures; a preparer who folds one in has changed what the figures measure without saying so, which the reviewer treats as a reportable error.

Where a quantity is measured at a state of the transaction rather than on a calendar date, the state is described in the terms the governing policy uses for it and the calendar date on which that state was reached is recorded alongside. Both are required: the state establishes what was measured and the date establishes when it can be reproduced from.

Section 4. Day-Count and Accrual Conventions

Interest, accretion and any other time-based amount accrues on the day-count basis the governing instrument designates for it. The preparer takes the basis from the instrument, records the clause it was taken from, and applies it without substitution; where the instrument is silent the preparer escalates rather than applying the basis used for a comparable instrument.

The bases this memorandum recognises are defined in the schedule to this section. A definition here is a definition only: it does not assign a basis to any instrument or any quantity, and a preparer may not read the order of the table as a preference. Where an instrument designates a basis this memorandum does not define, the preparer records the designation verbatim and asks the head of valuation to determine how it is to be applied.

Accrual is computed from and including the start date and to but excluding the end date unless the instrument provides otherwise, and the preparer states on the face of the computation which convention was applied. Where an instrument's own words settle the question the instrument governs, and the preparer cites the words rather than recording that the standard convention was used.

A model that accrues across a change in basis, a conversion, or an amendment must show the accrual in segments with the basis applied to each. A single accrual figure spanning a change of basis cannot be reproduced by a reader and is rejected at review even where the total is arithmetically correct.

An accrual is computed on the base the instrument identifies, and where that base itself changes during a period the model shows the base by segment. A preparer may not average a changing base across a period and may not accrue on the period-end base as a shortcut: both produce a figure the instrument does not provide for, and neither can be traced to a clause when a reviewer asks.

Compounding is applied only where the instrument provides for it, at the frequency the instrument states. Where an instrument is silent on compounding the preparer treats the amount as simple and records that the instrument was silent, rather than adopting the treatment used for a comparable instrument; the record is what allows the question to be revisited if counsel reads the clause differently.

Day counts are computed from dates, never from month arithmetic. A model that derives an accrual period by counting months and multiplying will diverge from the instrument in any period containing a month of unequal length, and the divergence grows with the term; the reviewer re-performs a sample of periods from the dates themselves.

Schedule 4.5 - Recognised day-count bases and their definitions

Basis	Definition	Point to watch	Assignment taken from
Actual/365 fixed	Actual days elapsed over a fixed denominator of 365	Denominator does not change in a leap year	Instrument terms
Actual/360	Actual days elapsed over a fixed denominator of 360	Produces more interest than a 365 denominator for the same period	Instrument terms
Actual/actual	Actual days elapsed over the actual days in the period year	Denominator changes in a leap year	Instrument terms
30/360 United States	Thirty-day months over a 360-day year, with the end-of-month adjustments the market convention applies	Adjustment rules must be applied in full, not selectively	Instrument terms
30E/360	Thirty-day months over a 360-day year on the European adjustment rules	Differs from the United States rule only in the end-of-month treatment	Instrument terms
Monthly compounding on stated periods	Periods defined by the instrument rather than by day count	Period boundaries taken from the instrument, not from a calendar	Instrument terms

Section 5. Precision, Rounding, and Order of Operations

Every quantity a model reports has a precision and a rounding direction, and both are taken from Schedule P for the quantity in question. The vocabulary of directions is defined in the schedule to this section; the assignment of a direction and a precision to a quantity is made only in Schedule P, and a preparer who cannot find the quantity there escalates rather than choosing.

Rounding is applied once, at the point the quantity is reported, and intermediate results are carried at full available precision. A model that rounds an intermediate result and then computes from the rounded value produces a figure that cannot be reproduced from the inputs, and the reviewer rejects it; where a governing instrument requires an intermediate quantity to be rounded, that requirement is cited and the rounding is shown as a separate step.

Order of operations must be explicit wherever it changes the result. Where a computation multiplies and divides quantities that are themselves rounded, the model shows the sequence in separate cells or separate lines rather than in a single compound expression, so a reviewer can see where each rounding was applied.

A reported quantity may not be presented at a precision finer than the precision at which it was computed, and a quantity computed at a finer precision may not be presented at a coarser one without the coarser presentation being identified as rounded for presentation. Both failures mislead a reader about how much the figure is capable of supporting.

Schedule 5.5 - Rounding direction vocabulary used by Schedule P

Direction	Definition	Typical application	Assigned in
Nearest, ties away from zero	The nearer representable value; an exact half moves away from zero	Amounts of money and stated rates	Schedule P
Nearest, ties to even	The nearer representable value; an exact half moves to the even digit	Statistical summaries	Schedule P
Toward zero	Truncation of the excess digits without adjustment	Quantities that may not be created by rounding	Schedule P
Away from zero	The next representable value in the direction of larger magnitude	Quantities that must be at least a computed minimum	Schedule P
Floor to a stated increment	The largest multiple of the increment not exceeding the value	Divisible quantities with a minimum increment	Schedule P
Ceiling to a stated increment	The smallest multiple of the increment not below the value	Requirements expressed as a minimum	Schedule P

Governing provision and cross-references

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This provision is read with the defined terms, the schedules to this instrument, and any consent or waiver recorded against it; a summary prepared from it does not vary it.

Section 6. Units, Scaling, and Presentation of Quantities

Every column and every line of a model output states its unit, and the unit is stated on the column rather than inferred from the values. Mixing units within a column is prohibited even where the values are distinguishable, and a model that presents a count and an amount in the same column is corrected before issue rather than annotated.

Scaling is declared and applied uniformly. Where an output is presented in thousands or millions the scale appears in the column heading and every value in the column is scaled; a single unscaled value in a scaled column is the most common cause of a misread output and the reviewer checks for it specifically.

Counts are presented as integers and are never scaled. A quantity that must be a whole number because the thing it counts is indivisible is computed so that the whole number falls out of the computation rather than being produced by presentation rounding, and the model shows where the adjustment to a whole number occurred.

Negative values are presented in one form throughout an output, and the form is declared. A model that uses parentheses in one schedule and a leading minus sign in another is corrected before issue, because a reader comparing the two schedules cannot tell whether the difference is presentational.

Section 7. Currency Translation and the Designated Rate Source

7.1 A quantity denominated in a currency other than the reporting currency of the fund is translated using the applicable rate from the source designated in the currency schedule to this memorandum, taken as at the date the schedule designates for the computation in question. The preparer records the source, the date and the rate applied on the face of the computation; a translated figure presented without those three items cannot be reproduced and is returned unread.

7.2 No other rate source may be used, and a rate obtained from a market data screen, a bank quotation or a counterparty's own statement may not be substituted for the designated source even where it is contemporaneous and even where the difference is immaterial. Where the designated source does not publish a rate for the pair or the date required, the preparer escalates to the head of valuation, who determines the treatment in writing and records the determination against the computation.

7.3 The rate is applied to the source amount, and the translated result is rounded once at the precision Schedule P assigns to the translated quantity. A preparer may not translate an already rounded intermediate figure, may not translate a total by applying the rate to the sum of differently dated components, and may not carry a translated figure into a later period without restating it if the schedule designates a different date for that period.

7.4 Both the source amount and the translated amount are retained in the model and in any output that reports the translated amount, together with the identification of the rate source. Where a figure passes through more than one model, each model retains the source amount rather than the translated amount alone, so that a later change in the designated source or date can be applied without rebuilding the chain.

7.5 A translation is a convention and not a judgement. Where a document under review states a translated figure that differs from the figure this section produces, the difference is recorded as a discrepancy under the document review protocol and reported; the preparer does not adopt the document's figure and does not adjust the firm's convention to reproduce it.

Section 8. Circular Calculations: Iteration, Convergence, and Termination

A calculation is circular where a quantity depends on itself through the calculation chain. Circular calculations are permitted where the dependency is real, and are resolved by iteration under this section rather than by a manual approximation, a plug, or a hard-coded value copied from a previous version of the model.

Iteration proceeds from a stated starting value until the change between successive iterations is within the tolerance the convergence schedule states for the quantity, or until the iteration ceiling that schedule states is reached. The starting value, the tolerance applied, the number of iterations performed and the final change are retained as the convergence evidence, and an output whose convergence evidence is absent is treated as not having converged.

Where the ceiling is reached without convergence the model does not report a result. The preparer reports the non-convergence, the sequence of iterates, and the preparer's analysis of why the calculation does not settle; the head of valuation determines whether the structure of the calculation is wrong or the tolerance is inappropriate, and records the determination.

A model that resolves a circularity by iteration must produce the same result on every run from the same inputs. Where the software's own iteration settings could affect the result, the settings are recorded with the model and the calculation is re-performed independently by a reviewer who sets them from the record rather than from the reviewer's own defaults.

Section 9. Tie-Break and Deterministic Resolution Rules

Where two or more results are equally supported by the conventions, the model must resolve between them by a stated rule rather than by the order in which the preparer happened to compute them. The rule is declared on the face of the model and produces the same outcome on every run; a resolution that depends on sort order, on the order rows were entered, or on an unstated preference is not a rule for this purpose.

Where an allocation must be made among holders and the arithmetic leaves a residual that cannot be divided at the required precision, the residual is allocated by the rule the governing instrument states. Where the instrument is silent, the preparer escalates; the residual may not be allocated to the largest holder, to the first row, or to a rounding account created for the purpose.

Where two governing documents would each supply a value for the same input, the document review protocol determines which governs, and the model cites the determination rather than making its own choice. A model that resolves a documentary conflict inside a formula has hidden the conflict from every subsequent reader and is rejected on that ground alone.

Every tie-break the model applies is listed in its documentation with the rule applied and the authority for it. The reviewer re-performs each listed tie-break, and an unlisted tie-break discovered during re-performance is reported as a control failure whether or not it changed the result.

A tie-break must be stable across runs and across data. A rule that resolves in favour of the first row resolves differently once a row is added, and a rule that depends on a sort resolves differently when the sort key contains duplicates; where a model must sort, it sorts on a declared key that is unique across the population and states the key in its documentation.

Where a tie-break would determine an amount payable to or receivable from a person, the model does not apply one. The preparer escalates, because a rule the firm has chosen for consistency is not an authority for allocating value between parties; the authority must come from a governing document, and where none exists the question is a legal one rather than a modelling one.

The register in the appendix to this memorandum lists the situations in which a tie-break is required and the rule each takes. It is not a menu of options: the preparer applies the rule listed for the situation and escalates where the situation is not listed. A situation that recurs unlisted is reported so the register can be extended rather than worked around case by case.

Section 10. Reconciliation Trail and Tie-Out Requirements

Every output a model issues must be reconcilable to its inputs and to the previously issued output, and the reconciliation is prepared as part of the issue rather than on request. The tie-outs in the schedule to this section are performed at the stated point by the stated person, and the evidence of each is retained with the version that was issued.

A movement between two issues is explained by input changes, not by a statement that the model was updated. Each explained movement identifies the input that changed, its previous and new value, and the document that supports the new value; an unexplained residual is disclosed as unexplained rather than absorbed into another line.

Where a figure is restated, the restatement shows the figure it replaces, the reason, and the date. The reviewer confirms that every register, report and model that carried the replaced figure has been notified, and the confirmation is retained; a restatement whose consumers have not been identified is incomplete.

Schedule 10.4 - Required tie-outs and the evidence each produces

Ref	Tie-out performed	When	Performed by	Evidence retained
TO-1	Capital structure lines to executed instruments	Before any decision use	Deal analytics	Instrument reference against each line
TO-2	Model inputs to the source document page	Before any decision use	Model owner	Page or clause citation per input
TO-3	Model outputs to the prior approved version	Every issue	Model owner	Movement explained by input change
TO-4	Currency-translated amounts to the source amount and designated rate	Every issue	Model owner	Both the source amount and the rate source recorded
TO-5	Iterative results to the convergence evidence	Every issue of an iterative model	Model owner	Iteration record retained
TO-6	Reserve model outputs to the reserve register	Every cycle	Reserve analyst	Register entry referenced
TO-7	Reporting model outputs to the accounting records	Every reporting period	Fund controller	Reconciliation statement filed
TO-8	Restated figures to the figure they replace	On every restatement	Model owner	Both figures shown with the reason

Section 11. Model Documentation, Review, and Change Control

A model is documented on its own face. The inputs sheet lists every input with its source document and the page or clause it was taken from; the conventions declaration states the schedule editions applied; the change log records every structural change with its date and its author. Documentation held separately from the model is not documentation for this purpose, because it is not what a later reader will find.

Review is performed by a person who did not build the model and consists of re-performing the calculation chain from the inputs, not of reading the formulas. The reviewer records what was re-performed and what was not, and the record accompanies the approval; an approval that does not state the scope of the re-performance is treated as unscoped and does not support decision use.

A structural change lapses the approval. The owner logs the change, the reviewer re-performs the affected chain, and the register records the new approval with its date; in the interval the model may not support a decision, and an output issued in that interval is withdrawn. A change to an input value alone does not lapse the approval but does require the tie-outs in section 10 to be re-performed.

The controls in the schedule to this section are the minimum, and the head of valuation may require more for a model whose output is load-bearing for a committee decision. A model that fails a control is not approved; there is no partial approval, and the register shows the control that failed rather than showing the model as pending.

Schedule 11.5 - Minimum model controls and the effect of failure

Control	Applies to	Evidenced by	Effect of failure
Inputs sheet separated from calculation	Every model	Model owner	No input value may appear inside a formula
Every input traced to a source document	Every model in decision use	Model owner	Untraced input blocks approval
Conventions declared on the face of the model	Every model	Model owner	Undeclared convention treated as unapplied
Independent re-performance of the calculation chain	Models in decision use	Reviewer independent of the owner	Difference reported before approval
Version retained for every issued output	Every model	Model owner	Output without a retained version withdrawn
Change log entry for every structural change	Every model	Model owner	Unlogged change invalidates approval

Control	Applies to	Evidenced by	Effect of failure
Convergence evidence retained for iterative calculations	Iterative models	Model owner	Absence treated as non-convergence
Reconciliation of outputs to the accounting records	Reporting models	Fund controller	Unexplained difference escalated
Approval recorded on the model register	Models in decision use	Head of valuation	Unapproved model may not support a vote

Section 12. Defined Terms, Schedules, and Version History

Convention means a rule stated in this memorandum. Assignment means the allocation of a particular precision, direction, basis, source, tolerance or ceiling to a particular quantity, made only in the schedules this memorandum names. Decision use means use to support a committee decision, and reporting use means use to produce a figure issued outside the firm.

Schedule P holds precision and rounding assignments; the currency schedule holds the designated rate source and the dates at which rates are taken; the convergence schedule holds tolerances and iteration ceilings; the measurement calendar holds designated reporting dates. Each schedule is amended by the head of valuation with the approval this memorandum requires, and each carries its own edition and effective date.

A reference to a schedule is to the edition in force when the computation was performed, and the computation records that edition. Where a schedule is amended after a computation was performed but before its output is used, the output is re-performed on the amended schedule or is reissued with the edition it applied stated on its face; it may not be used silently under the earlier edition.

Schedule 12.4 - Issue history of the modelling conventions memorandum

Issue	Approved	Applies from	Change on issue	Standing
Issue 2	2019-02-26	2019-03-18	First consolidated conventions memorandum	Withdrawn
Issue 3	2021-05-11	2021-06-01	Precision vocabulary separated from assignments	Withdrawn
Issue 3 rev 2	2022-08-30	2022-09-19	Iteration and convergence conventions added	Withdrawn
Issue 4	2024-04-23	2024-05-13	Currency section restated; rate source designation moved to the schedule	Withdrawn
Issue 4 rev 2	2025-09-09	2025-09-29	Tie-break rules made explicit and testable	Withdrawn
Issue 4 rev 3	2026-03-24	2026-04-06	Current issue; reconciliation trail requirements extended to overlays	Current

Appendix A. Order in which the conventions are applied to a computation

The conventions in this memorandum are applied in the order set out in the schedule below, and the order matters: a preparer who translates before establishing the measurement date, or who rounds before completing the calculation chain, will produce a figure that cannot be reproduced by a reviewer following the order. The schedule states the step, what must be settled before it, and the record the step produces.

The order is not a suggestion about workflow. It is the sequence in which a reviewer will re-perform the computation, and a model built in a different sequence will not survive re-performance even where its arithmetic is sound. Preparers building a new model should lay the model out in this order, one group of sheets to a step, so the re-performance follows the sheets.

Where a step does not apply to a computation, the record for that step states that it does not apply and why. An omitted step and an inapplicable step look identical to a reviewer, and the distinction is frequently the whole of the question: a computation with no translation step either involves one currency or has failed to translate.

A preparer who cannot complete a step escalates at that step rather than continuing on an assumption and returning to it. Work performed after an unresolved step has to be redone when the step resolves, and the practice of pressing on has produced more rework at this firm than any other single habit.

Schedule A.1 - Application order, prerequisites and the record each step produces

Step	What is done	Must be settled first	Record produced
A	Identify the quantity and the question it answers	Nothing	Statement of the quantity on the inputs sheet
B	Establish the measurement date or state measured	Step A	Date or state recorded with its source
C	Identify the governing document for each input	Step A	Input list with clause or page citations
D	Take the designated basis, source and precision	Steps B and C	Conventions declaration naming schedule editions
E	Translate amounts to the reporting currency	Step D	Source amount, rate source, date and result
F	Compute the chain at full available precision	Step E	Calculation sheets with no embedded inputs
G	Resolve any circularity by iteration	Step F	Iteration record for each circularity
H	Apply any tie-break the situation requires	Step G	Tie-break entry citing its authority
I	Round once at the reporting precision	Step H	Rounded output with the direction named
J	Perform the required tie-outs	Step I	Tie-out evidence for the version issued
K	Obtain review and record its scope	Step J	Reviewer's re-performance record
L	Register approval and issue	Step K	Model register entry with the approval date

Appendix B. Modelling defects this firm has shipped, and what prevents each

The defects below are recorded because the firm produced them, not because they are theoretically possible. Each entry states the defect, the failure it causes and the practice that prevents it, and each was added after a review or a reconciliation found it in an issued output. The catalogue is amended when a new defect is found rather than on a review cycle, so that it remains a record of experience.

Reviewers work through this catalogue on the first review of a model and record that they have done so. The record is part of the approval: a reviewer who has not worked through the catalogue has performed a narrower review than this memorandum requires, and the register shows the scope the reviewer actually covered rather than a general statement that the model was reviewed.

Two of these defects are invisible in the output and can be found only in the file. An input typed inside a formula and a value copied from a previous version both produce a figure that looks correct and that will not update when its source changes; they are found by reading the calculation, which is why the review requires re-performance from the inputs rather than a review of the results.

A defect found in an issued output is reported to the head of valuation with the output, the versions affected, and the consumers who were notified. The report is made whether or not the defect changed a decision, because the catalogue's value depends on the firm recording defects that turned out to be harmless as well as those that did not.

Schedule B.1 - Catalogue of modelling defects and preventing practice

Defect observed	Failure it causes	Practice that prevents it	Found by
Input value typed inside a formula	The input cannot be found or changed by a reviewer	One row per input on the inputs sheet	Re-performance from inputs
Intermediate result rounded and then reused	The output cannot be reproduced from the inputs	Round once, at the reporting point	Re-performance from inputs
Value copied from a previous version of the model	The figure does not update when its source changes	Every value either an input or a formula	Reading the calculation
Figure reported without its date or state	The reader cannot tell what was measured	Date on the face of every output	Review of the output
Scale declared only in a footnote	A reader scanning columns misreads every value	Scale in the column heading	Review of the output
Accrual spanning a change of basis shown as one figure	The basis applied cannot be identified	Segment the accrual and label each part	Tie-out to the instrument
Circularity resolved by a manually entered plug	The dependency is hidden and will not update	Iterate and retain the convergence record	Reading the calculation
Translated figure carried forward without its source amount	A change of rate source cannot be applied	Retain the source amount in every model	Currency tie-out
Documentation held outside the	A later reader does not find it	Document on the face of the	Handover of ownership

Defect observed	Failure it causes	Practice that prevents it	Found by
model file		model	
Approval carried across a structural change	The approved calculation is not the one in use	Log the change and re-approve	Register reconciliation
Unexplained residual absorbed into another line	The movement analysis becomes untestable	Disclose the residual as unexplained	Movement analysis review
Total adjusted so reported components add	A component is misstated to preserve an appearance	Disclose the rounding difference	Re-performance of the total

Appendix C. Schedules relied on, competence, and attestation

The schedules indexed below hold every assignment this memorandum applies. Each is maintained by the named function, carries its own edition and effective date, and is amended only with the approval this memorandum requires. A preparer records the edition of each schedule relied on against the computation, and a computation that names no edition is treated as having applied none.

A preparer may build or amend a model in decision use only while holding a current attestation of familiarity with the issue of this memorandum in force. The attestation is renewed on each issue rather than annually, because the issues are what the attestation is about; the head of valuation maintains the register of attestations and reports lapses to the committee.

A reviewer approving a model must hold the same attestation and must additionally have re-performed at least one model of the class within the preceding period. Approval by a reviewer who has not is withdrawn on discovery and the model is re-reviewed; the register records the withdrawal so the effect of the competence requirement is visible rather than assumed.

Where a computation is prepared by an adviser, the engagement letter requires the adviser to apply these conventions and to state the schedule editions applied. The firm does not accept an adviser's own conventions in place of these, and a deliverable that applies them is returned; where the adviser's professional standards require a different treatment, the difference is identified in writing before the work begins.

Schedule C.1 - Index of schedules, maintainers and review cadence

Schedule	What it holds	Maintained by	Reviewed	Applies to
Schedule P	Precision and rounding assignments by quantity	Head of valuation	On amendment	Every model
Currency schedule	Designated rate source and the dates rates are taken at	Head of valuation	On amendment	Any translated quantity
Convergence schedule	Tolerances and iteration ceilings by quantity	Head of valuation	On amendment	Iterative models
Measurement calendar	Designated reporting dates	Fund controller	Annually	Reporting models
Instrument conventions schedule	Where each instrument's designated basis is recorded	Deal analytics	On each instrument	Accrual computations
Model register	Class, owner, approval and permitted use of each model	Head of valuation	Continuous	Every model
Change log standard	The fields a model change log must carry	Head of valuation	On amendment	Every model
Tie-out standard	The tie-outs required before an output is issued	Fund controller	On amendment	Every issued output

Schedule C.2 - Situations requiring a tie-break and the rule each takes

Ref	Situation	Rule applied	Recorded as
TB-1	Two documents supply the same input	Apply the review protocol determination	Determination cited in the model
TB-2	Residual after allocation at the required precision	Apply the rule the governing instrument states	Clause cited; escalated if silent
TB-3	Two measurement dates designated for one quantity	Escalate; do not select	Escalation recorded with both dates
TB-4	Equal-ranking results from two permitted methods	Apply the method the schedule designates	Schedule edition recorded
TB-5	Rounding produces two representable values	Apply the direction Schedule P assigns	Direction named on the output
TB-6	Sort order would change an allocation	Sort on a declared key unique across the population	Key declared in the documentation
TB-7	Two instruments designate	Segment the accrual	Segments shown separately

Ref	Situation	Rule applied	Recorded as
	different bases for one accrual		
TB-8	A rate is unavailable for a designated date	Escalate to the head of valuation	Written determination filed
TB-9	An input exists in two units	Convert once, on the inputs sheet	Conversion shown on the inputs sheet

Appendix D. Sensitivities, scenarios, and illustrative figures

A sensitivity varies one input of an approved model and reports the effect; a scenario varies a coherent set of inputs to describe a state of the world. The two are labelled differently because they support different arguments, and a preparer who labels a scenario as a sensitivity invites a reader to treat a set of assumptions as a single measured effect.

An overlay is built on an approved base model and never replaces it. The base model's inputs are not edited; the overlay carries the varied inputs, cites the base model and its version, and reports the base result alongside the varied one. An overlay whose base result cannot be reproduced from the cited version is withdrawn, because the comparison is the whole of what an overlay reports.

Every varied input is named with the value it was varied to and the reason the variation was chosen. A range presented without the reason its endpoints were selected tells a reader nothing about how likely either endpoint is, and the firm has found that unexplained ranges are read as confidence intervals when they are nothing of the kind.

An illustrative figure is one produced to show how a mechanism works rather than to state what a quantity is. It is labelled as illustrative on the face of the output, at every place it appears, and it may not be carried into a register, a report or another model. A figure that has escaped its label is treated as a misstatement rather than a presentational lapse.

Overlays are retained with the base model and are subject to the same version discipline, but they are not approved for decision use in their own right. Where a committee decision would rest on an overlay result, the overlay is promoted to a model of its own class, registered, reviewed and approved; the shortcut of deciding on an unregistered overlay is the failure this rule exists to prevent.

A sensitivity that shows a decision changing within the range of a single input is reported to the committee explicitly rather than left to be noticed. The preparer states which input, the value at which the decision changes, and what evidence supports the input's value; a decision that turns on one input is a different kind of decision from one that does not.

Scenario labels are drawn from the set the head of valuation maintains, so that a scenario described in one paper means the same thing in the next. A preparer who needs a scenario the set does not contain proposes an addition rather than coining a label, and the proposal states what distinguishes the new scenario from those already defined.

No overlay may be circulated outside the firm without the base model result and the identification of the varied inputs. The firm has recalled circulated overlays for this reason, and the recall is recorded on the model register so the pattern is visible.

Schedule D.1 - Overlay rules and the failure each prevents

Rule	Applies to	Failure it prevents	Evidenced by
Base model cited by name and version	Every overlay	A comparison against an unknown base	Citation on the overlay
Base result reported alongside the varied result	Every overlay	A varied figure read as a measurement	Both results on the output
Varied inputs named with their values	Every overlay	An unattributable difference	Varied input list
Reason for each endpoint stated	Every range	A range read as a confidence interval	Reason recorded per endpoint
Sensitivity and scenario labelled distinctly	Every overlay	A set of assumptions read as one effect	Label on the output
Scenario label drawn from the maintained set	Every scenario	The same label meaning different things	Set reference on the output
Illustrative figures labelled at every appearance	Every illustration	An illustration entering a register	Label beside each figure
Overlay not used for a decision unless registered	Every overlay	A decision on an unreviewed calculation	Model register entry
Decision-changing input reported explicitly	Every sensitivity	A fragile decision taken as robust	Statement in the paper

Rule	Applies to	Failure it prevents	Evidenced by
External circulation includes the base result	Every overlay	A varied figure circulating alone	Circulation record

Schedule D.2 - Labels available for an overlay and what each asserts

Label	What it asserts	What it does not assert
Sensitivity	The effect of varying one named input	That the varied value is likely
Scenario	A coherent set of inputs describing one state	That the state is the expected one
Downside case	A state the preparer considers adverse and plausible	A floor on the outcome
Upside case	A state the preparer considers favourable and plausible	A ceiling on the outcome
Break-even analysis	The input value at which a stated outcome changes	That the input can reach that value
Illustration	How a mechanism operates	Any quantity of the transaction
Stress case	A state chosen to test whether a conclusion survives	That the state is expected at all